Research Article Vol. 16, No. 4 / 1 Apr 2026 / Optical Materials Express 1054
Optical Materials EXPRESS

Influence of annealing temperatures on the nonlinear optical performance of fluorine-doped tin oxide thin film

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Abstract: Fluorine-doped tin oxide (FTO) thin films exhibit potential applications in photonic devices, owing to their superior transparency, conductivity, and adjustable nonlinear optical response. In this study, FTO films were prepared through magnetron sputtering, and the influence of annealing temperature on their crystalline structure and third-order nonlinear optical properties was systematically investigated. The results indicate that annealing enhances the carrier concentration from $0.93 \times 10^{20}\,cm^{-3}$ to $1.03 \times 10^{20}\,cm^{-3}$. Z-scan measurements confirm that the film annealed at 700 °C exhibits optimal nonlinear parameters with a nonlinear absorption coefficient of -5.7×10^{-9} m/W and a nonlinear refractive index of $6.9 \times 10^{-16}\,m^2/W$. The third harmonic generation conversion efficiency increases with annealing temperature. This research clarifies the mechanism driven by carrier dynamics that underlies the annealing-induced nonlinear enhancement in FTO, and offers a robust materials foundation for the development of ultrafast all-optical modulators, high-sensitivity nonlinear sensors, and on-chip frequency converters.

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1. Introduction

Transparent conducting oxides (TCOs) integrate high optical transparency and excellent electrical conductivity, thus becoming indispensable in optoelectronic and photonic technologies [1–3]. Beyond their linear optical behavior, TCOs have recently garnered increasing attention as CMOS-compatible platforms for nonlinear photonics, where carrier-dominated dynamics can enable ultrafast all-optical modulation, ultrafast optical sensing and frequency conversion [4,5].TCOs comprise a broad class of material systems, with representative examples including indium tin oxide (ITO), aluminum-doped zinc oxide (AZO), and fluorine-doped tin oxide (FTO). ITO is currently the most widely used TCO, and its nonlinear optical properties and tunability have attracted considerable attention; notably, ITO has been demonstrated to be of practical value for silicon photonics and related integrated-photonic applications [5–9]. However, ITO also faces notable limitations: indium is scarce and expensive, and ITO is considered toxic. Consequently, it is necessary to explore alternative TCO materials. Among the various TCOs, FTO is especially attractive owing to its excellent thermal stability, mechanical durability, and cost-effectiveness compared with indium-based oxides [10,11]. Despite numerous studies on the linear optical and electrical properties of FTO films, systematic explorations of the nonlinear optical responses of FTO remain limited, particularly at telecommunication wavelengths.

#590743 https://doi.org/10.1364/OME.590743
Journal © 2026 Received 19 Jan 2026; revised 15 Mar 2026; accepted 16 Mar 2026; published 24 Mar 2026

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Effective tuning of nonlinear optical effects in TCOs is crucial for their practical application in integrated photonic devices. For TCO materials, tuning of the nonlinear optical response has been realized through various strategies, including chemical doping-based carrier concentration engineering [12], oxygen vacancy tuning [13,14], electrostatic gating [15], and field enhancement effects based on artificial micro structures or hybrid multilayer stacks [16,17]. However, these methods often suffer from problems such as increased optical losses, enhanced fabrication complexity, and limited scalability. In contrast, magnetron sputtering technology offers a more direct and scalable solution, enabling tuning of carrier concentration during thin-film deposition, thereby achieving systematic modulation of the nonlinear response [18]. By adjusting deposition parameters, including sputtering power, working pressure, oxygen partial pressure, and substrate temperature, in conjunction with post-deposition annealing conditions, the free-carrier concentration, scattering and the populations of defects such as oxygen vacancies can be simultaneously modified. Consequently, the Drude response and linear optical dispersion of the material [13,19] can be reconfigured. Simultaneously, the crystalline microstructure of the film encompassing grain size, texture, crystallinity, and interface quality, exerts further influences on carrier transport and optical loss [20]. Research has also shown that the third-order nonlinearity of TCOs is jointly determined by carrier density and microstructural quality. The magnitude and even the sign of nonlinear absorption and refraction can undergo significant alterations with processing conditions, and the harmonic-generation efficiency is correspondingly impacted [20–24]. These studies provide the foundation for the tunability of nonlinear optical properties. Nevertheless, systematic studies on FTO are still relatively scarce, especially at telecommunication wavelengths. Moreover, correlation between the annealing-induced alterations in carrier concentration and crystallinity and nonlinear parameters is still absent.

In this research, we present a comprehensive experimental investigation into the influence of annealing temperature on the structural, electrical, and third order nonlinear optical properties of FTO thin films. The films were deposited onto quartz substrates via RF magnetron sputtering and subsequently annealed at temperatures ranging from 500 to 800 °C in an air environment. Structural and chemical characterizations were carried out using X-ray diffraction (XRD) and X-ray photoelectron spectroscopy (XPS), whereas Hall effect measurements were employed to determine the carrier concentration and resistivity. Optical transmittance spectra were measured for analyzing the linear absorption property. The third order nonlinear optical parameters were measured by the single beam Z-scan technique with a femtosecond laser, and third harmonic generation (THG) experiments were conducted to evaluate the nonlinear conversion efficiency. The results indicate that annealing temperature can modulate the crystallinity and carrier concentration, which consequently improves the nonlinear refractive index and THG intensity. This study can establish a correlation between crystal structural, carrier density, and the third order nonlinear optical response.

2. Results and discussions

2.1. FTO thin film preparation and characterization

FTO thin films were deposited on pre-cleaned glass substrates through magnetron sputtering. A fluorine-doped SnO_2 ceramic target (95:5 wt% SnO_2:F, 50.8 mm diameter, 3 mm thickness) was employed under Ar atmosphere [25]. The working pressure, power was maintained constant to guarantee uniform film growth. Figure 1(a) depicts a schematic representation of the sputtering process for fabricating FTO thin films. During the deposition process, no intentional substrate heating was implemented. Soda-lime glass substrates were selected due to their excellent adhesion to SnO_2 and their capacity to endure annealing temperatures up to 800 °C without deformation. The schematic diagram for FTO film deposition is presented in Fig. 1(a). Annealing is extensively acknowledged to enhance crystallinity, diminish defect density, and relieve internal stress in oxide films [26,27]. In this study, systematic annealing was utilized to optimize the morphology,

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optical transparency, and structural uniformity of FTO thin films. Post-deposition annealing was conducted at a temperature range of 500–800 °C to investigate the thermal impacts on film quality.

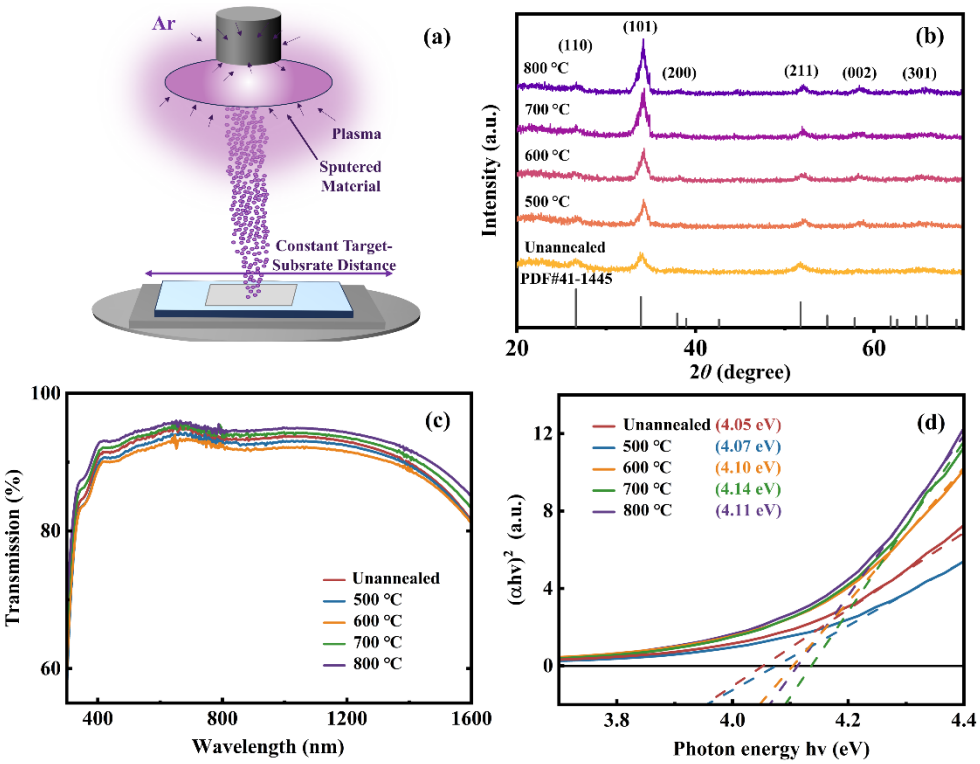


Fig. 1. (a) Schematic diagram of the magnetron sputtering setup for FTO film deposition; (b) the XRD patterns, (c) optical transmittance spectra and (d) Tauc plots of the FTO thin films annealed at different temperatures.

Figure 1(b) depicts the XRD patterns of FTO thin films annealed at different temperatures. All the diffraction peaks can be indexed to a tetragonal SnO_2 phase (JCPDS No. 41-1445). No secondary phases associated with fluorine doping were detected, which can be ascribed to the low doping concentration and the detection limitations of XRD [26]. The diffraction patterns limitations characteristic peaks corresponding to the (110), (101), (211), (002), and (301) planes of SnO_2, with the (101) peak being the most prominent, suggesting a preferred orientation [26,28]. The diffraction intensity, especially that of the (101) plane, increases with annealing up to 700 °C, indicating improved crystallinity and grain growth [26–29]. The diffraction peaks remained almost unchanged between 700 °C and 800 °C.

Figure 1(c) shows the optical transmittance spectra of FTO thin films. FTO films exhibit high transparency with an average transmittance exceeding 80% during 300–1600 nm region, regardless of the annealing temperature. After annealing, a slight blue shift in the optical bandgap is detected, from 4.05 eV to 4.14 eV, which is shown in Fig. 1(d). These results suggest that moderate annealing can tune the band structure.

Figure 2(a) shows the XPS survey spectrum of the FTO thin film in the binding energy range of 0–800 eV. The peaks of Sn 3d, Sn 4d, O 1s, F 1s, and C 1s are clearly identified. The Sn and O peaks confirm that the main composition is SnO_2, while the F 1s peak verifies successful fluorine doping. The C 1s peak arises from surface carbon contamination and was

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used for charge correction at 284.8 eV. The absence of impurity peaks indicates that the film has high purity and good stoichiometry [30]. Figures 2(b–f) present the O 1s high-resolution spectra of FTO films before and after annealing at 500–800 °C. Each spectrum can be fitted into two components located at about 530.8 eV and 532.0 eV, corresponding to lattice oxygen and chemisorbed oxygen or oxygen-vacancy-related species, respectively [31]. The proportion of the higher-binding-energy peak increases from 18.67% of unannealed to 45.57% at 700 °C, indicating the formation of more oxygen vacancies and free carriers [31,32]. At 800 °C, this proportion slightly decreases to 43.97%, suggesting partial reoxidation of the surface. Overall, the oxygen-vacancy concentration first increases and then slightly decreases with temperature, reflecting a competition between oxygen desorption and reoxidation. The 700 °C annealing condition provides the optimal balance between crystallinity, oxygen-vacancy concentration, and electrical conductivity.

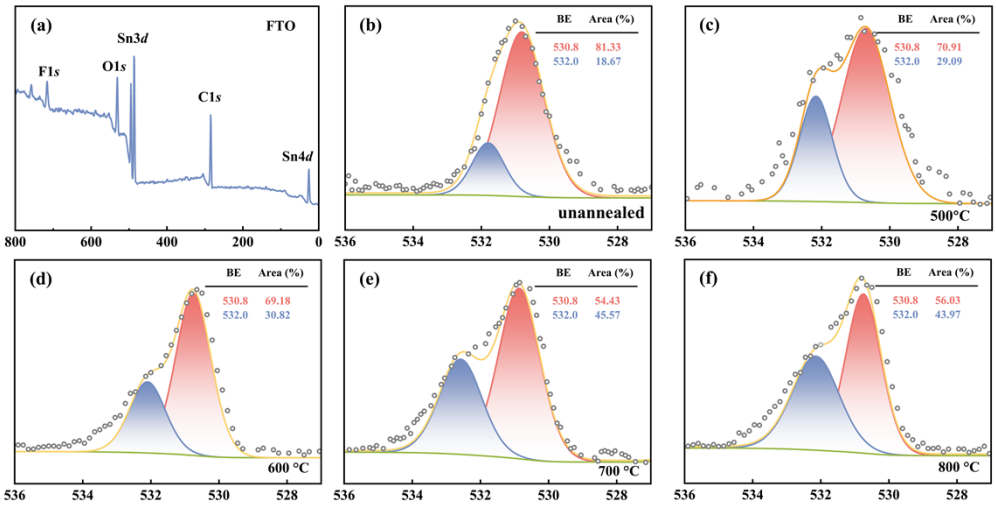


Fig. 2. (a) XPS survey spectrum of the surface layer of FTO thin films and O 1s spectra of FTO thin films annealed at different temperatures: (b) unannealed, (c) 500 °C, (d) 600 °C, (e) 700 °C, and (f) 800 °C

2.2. *Nonlinear optical properties*

To explore the impact of annealing temperature on the third-order nonlinear optical properties of FTO thin films, single beam Z-scan measurements were conducted. The schematic diagram of the Z-scan system in the experiment is shown in Fig. 3(a). The laser source was an optical parametric amplifier (OPA) pumped by a Ti: sapphire regenerative amplifier system (Astrella-USP, Coherent) operating at a 1kHz repetition rate and 35 fs pulse duration. By translating it along the optical axis of the focused beam, the sample was exposed to different pump intensities and the nonlinear optical response was investigated by recording the transmitted light. To improve accuracy, we employed a three-channel Z-scan configuration, in which an input reference channel and the transmitted signals were recorded simultaneously, enabling normalization of the measured transmittance and suppression of common-mode laser fluctuations. The beam diameter was approximately 140 µm, corresponding to a peak intensity of 89 GW/cm^2. The nonlinear measurements were performed at 1550 nm, a representative wavelength in the telecommunication C-band, to establish a basis for evaluating nonlinear materials and their compatibility with integrated photonic platforms.

The open-aperture Z-scan measurements of FTO thin films under different annealing conditions are depicted in Fig. 3(b). All samples exhibit a symmetric peak centered at the focus, which is a

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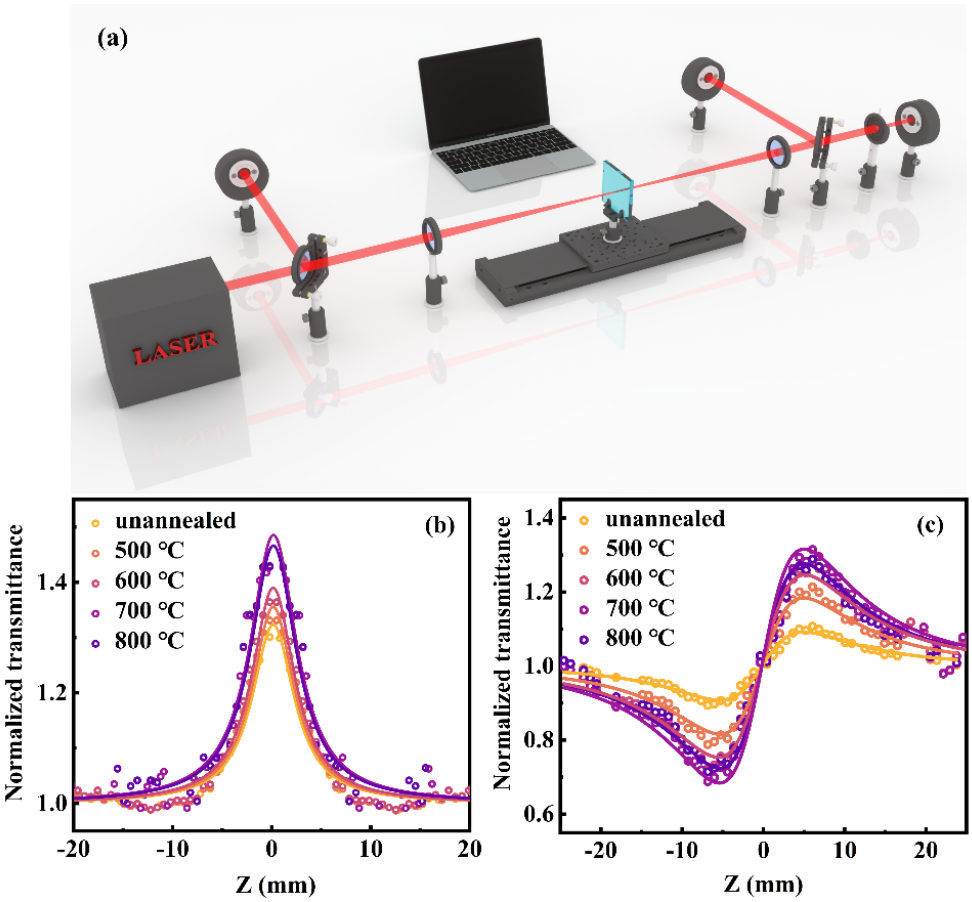


Fig. 3. (a) Schematic diagram of the Z-scan experimental setup, (b) Nonlinear absorption and (c) nonlinear refraction characteristics of FTO thin films annealed at different temperatures.

characteristic feature of saturable absorption behavior. The normalized transmittance gradually increases with the rise of annealing temperature, indicating an enhancement of the nonlinear absorption effect. Notably, the film annealed at 700 °C exhibits the most obvious transmittance variation, suggesting that the strongest nonlinear absorption at 700 °C annealing [33–35]. Figure 3(c) displays the closed-aperture Z-scan curves of the FTO films. The peak–valley curves correspond to a positive nonlinear refractive index. Similar to the nonlinear absorption response, the peak to valley difference increases with annealing temperature until reaching 700 °C, indicating the maximum nonlinear refraction is attained. This enhancement can be attributed to the improved carrier density, which enhance the local field and nonlinear interaction [20,34].

Nonlinear fitting was utilized to extract the nonlinear coefficients of the FTO thin film, as shown in Fig. 4. The maximum nonlinear absorption coefficient was -5.7×10^{-9} m/W, representing a fourfold increase over the value for the unannealed sample. Simultaneously, the nonlinear refractive index of 6.9×10^{-16} m^2/W, signifies a two times enhancement. To obtain the most accurate results possible, we excluded any nonlinear contribution from the substrate and then improved the reliability of the data by repeating the measurements multiple times. Considering the repetition rate of 1 kHz of the laser and the high optical transparency of the FTO film at the excitation wavelength, we conclude that thermal effects in the Z-scan measurements are negligible under the present experimental conditions. Nevertheless, a systematic investigation of possible

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thermal contributions under higher average-power or different operating regimes will be of interest in future work. These results further confirm that an optimal annealing temperature refines the microstructure and electronic characteristic of the FTO thin film, consequently enhancing its third order nonlinear response. Therefore, annealing temperature serves as a crucial parameter for customizing the nonlinear optical performance of FTO thin films.

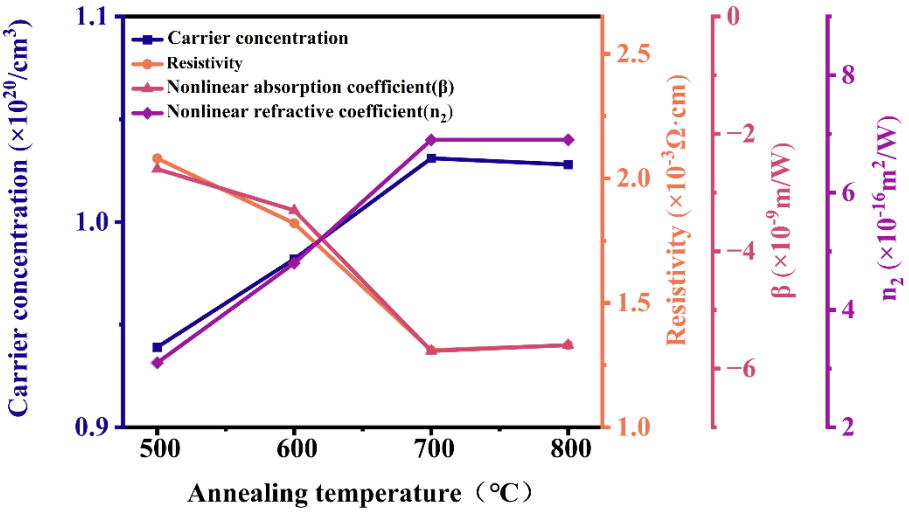


Fig. 4. Correlation between carrier concentration, resistivity, and nonlinear optical coefficients of FTO thin films annealed at different temperatures.

To explore the correlation between annealing temperature and the nonlinear optical properties, the electrical properties of FTO thin films annealed at different temperatures were characterized via Hall effect measurements. Figure 4 depicts the corresponding variations in carrier concentration and resistivity as functions of annealing temperature. When the annealing temperature was increased from 500 °C to 700 °C, the carrier concentration exhibited a steady increase, whereas the resistivity showed a substantial decrease. At 700 °C, optimal electrical performance was attained, with the carrier concentration to be 1.03×10^{20} cm^{-3} and the resistivity of 1.31×10^{-3} Ω·cm. A further increase in the annealing temperature to 800 °C led to a marginal decline in carrier concentration and a corresponding increase in resistivity. The degradation of electrical properties beyond 700 °C is probably ascribable to the reincorporation of oxygen from the ambient atmosphere at high temperatures. This phenomenon partially compensates for the oxygen vacancies, which serve as donor sources, consequently reducing the net carrier density. In the temperature range of 500–700 °C, the annealing process promotes the effective activation of fluorine dopants and the generation of oxygen vacancies, both of which act as donors and increase the free-electron density. Meanwhile, the improved crystallinity and reduced grain-boundary density suppress carrier scattering, thereby enhancing mobility and subsequently reducing resistivity [36]. At higher temperatures (>700 °C), oxygen incorporation from the surrounding environment partially compensates for the oxygen vacancies, decreasing the free-carrier density and increasing resistivity [31].

More fundamentally, the nonlinear susceptibility exhibits a strong dependence on carrier density [34,37]. In the near-infrared regime, the nonlinear behavior of FTO is predominantly governed by free-carrier effects [3,34,38]. Higher carrier densities enhance the free-carrier contribution to $\chi^{(3)}$, resulting in more pronounced nonlinear refraction and saturable absorption responses, as shown in Fig. 4. As a result, the annealing temperature functions as an effective parameter for adjusting the third-order nonlinear susceptibility of FTO thin films, thus regulating their nonlinear optical

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response. It is further demonstrated that an annealing temperature of approximately 700 °C offers the optimal condition, which balances crystallinity, carrier concentration, and conductivity to generate the strongest nonlinear optical response.

To further verify the nonlinear optical behavior of the FTO films, third-harmonic generation measurements were performed under femtosecond excitation at 1550 nm, as shown in Fig. 5. The experimental setup is similar to that described in our previous work [39]. As shown in Fig. 5(a), a distinct THG emission peak around 517 nm was observed for all films, and its intensity varied significantly with annealing temperature. As the annealing temperature increased, the THG signal gradually strengthened and reached its initial maximum at 700 °C, consistent with the Z-scan results. This indicates that annealing improves the film's crystallinity and carrier concentration, thereby enhancing the third-order nonlinear susceptibility [20,34]. When the annealing temperature was further raised to 800 °C, the THG intensity increased dramatically, exceeding that of the 700 °C sample. This differs slightly from the Z-scan results, where β and n_2 reached their maxima at 700 °C, suggesting that, in addition to carrier effects, phase matching and microstructural uniformity at higher annealing temperatures also play important roles in enhancing harmonic generation efficiency.

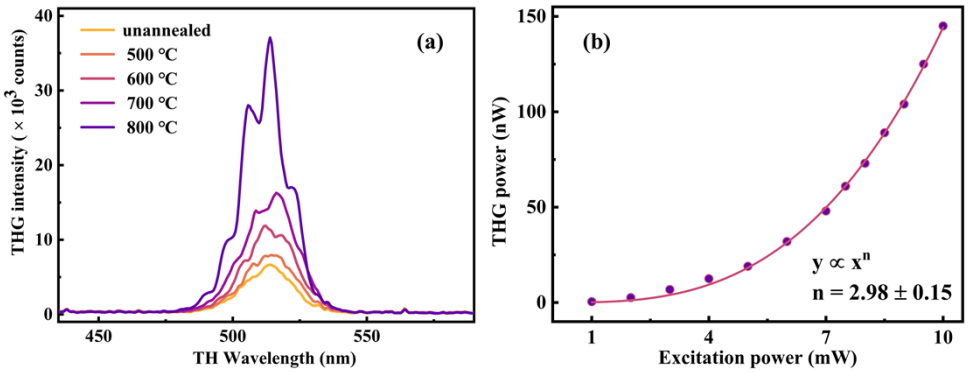


Fig. 5. Third-harmonic generation characteristics of FTO thin films under 1550 nm femtosecond excitation: (a) THG spectra of FTO thin films annealed at different temperatures under 1550 nm femtosecond laser excitation and (b) Power dependence of the THG signal measured for the 700 °C sample.

According to third-order nonlinear polarization theory, the third-harmonic intensity scales as: $I_{3\omega} \propto |\chi^{(3)}|^2 I_{\omega}^3$. Figure 5(b) presents the power-dependent THG characteristics of the 700 °C sample. The THG intensity exhibits a clear cubic dependence on the pump intensity, confirming that the signal originates from a third-order nonlinear optical process.

As summarized in Table 1, the annealed FTO films demonstrate superior wavelength-conversion performance relative to established materials. The THG efficiency of 1.45×10^{-5} for FTO exceeds that of silicon and ITO nanofilms by one to three orders of magnitude, while remaining comparable to AZO and CdO: In/Au films. These findings establish FTO as a highly promising material platform for high-efficiency third-harmonic generation and ultrafast nonlinear photonic applications.

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Table 1. Comparison of third-harmonic generation efficiency of FTO thin film with other reported optical materials

Optical Material	Thickness (nm)	Wavelength (nm)	THG Efficiency	Ref.
Silicon membrane	200	1550	6×10^{-9}	[40]
ITO film	33	1400	3.3×10^{-6}	[41]
AZO film	203	1550	0.63×10^{-6}	[42]
CdO: In/Au film	75/200	2100	10^{-5}	[43]
FTO film	180	1550	1.45×10^{-5}	This work

3. Conclusion

In summary, FTO thin films were prepared by RF magnetron sputtering and annealed at 500–800 °C to investigate the influence of thermal treatment on their structural, electrical, and third-order nonlinear optical properties at 1550 nm. XRD and XPS analyses confirmed enhanced crystallinity and moderate oxygen vacancy generation at 500–700 °C. Hall-effect measurements revealed a significant increase in carrier concentration and reduced resistivity with annealing. Z-scan measurements demonstrated that both nonlinear refractive index and THG efficiency increased with annealing temperature up to 700 °C, while β decreases, indicating improved film quality and reduced defect states. The correlation between nonlinear optical response and carrier density confirms that free carriers dominate the nonlinear response in the near-infrared regime. These findings provide a clear understanding of how annealing influences the interplay between microstructure, carrier dynamics, and third-order nonlinearity response characteristic in FTO films, suggesting a promising approach for designing low-loss, tunable nonlinear materials for near-infrared photonic and optoelectronic applications such as all-optical modulation and frequency conversion.

Funding. National Natural Science Foundation of China (62465015); Natural Science Foundation of Hunan Province (2023JJ30165, 2025JJ60404).

Acknowledgement. This work was supported by the National Natural Science Foundation of China (Grants No. 62465015); Natural Science Foundation of Hunan Province (2023JJ30165, 2025JJ60404).

Disclosures. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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